

A THESIS TYPESET WITH TYPST

The Faculty of Natural Sciences
University name for
the doctoral degree

Doctor rerum naturalium (Dr. rer. nat.)

submitted by

JOHN DOE

born in

SMALLVILLE

To XYZ
A quotation

ACKNOWLEDGEMENTS

I thank coffee and my friends. In that order.

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1.1 FIGURES

In this template, we have figures, as shown in Fig. 1.

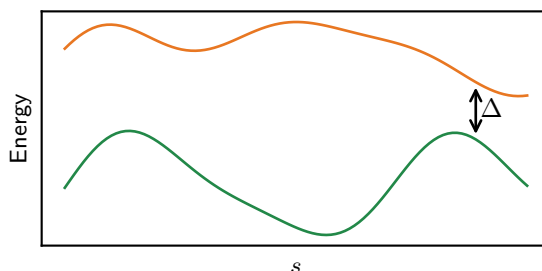


Fig. 1: An example figure. The caption font is slightly smaller than the body font.

1.2 CITATIONS

We also have nifty citation tools, for instance `@mike_and_ike[!]` will render as a prose citation, Nielsen and Chuang [1]. We can also add specifications, with `@mike_and_ike[!: p. 100]`, rendering as Nielsen and Chuang [1, p. 100]. The normal citation `@mike_and_ike[p. 100]` works as well: [1, p. 100]. Citations use two custom styles defined in `csl/`. The prose citation uses `short_and_sweet.csl`, while the bibliography section uses `thoughtful_and_coolheaded.csl` plus some show rules.

1.3 REFABLE-BOXES

In order to implement theorems and such, there's a higher-level function called `refable-box` (in the sense that it can be referenced). For example the theorem is implemented as

```
#let theorem(body) = refable-box(body, kind: "Theorem")
```

which gives

Theorem 1

Peach-flavored iced tea is superior to lemon-flavored iced tea.

Theorems also usually are followed by proofs, which can be called right away with `#proof(body)`

Proof: It's obvious.

1.4 ACRONYMS

Acronyms can be defined in the `acrodictionary` array at `inputs.typ`. Then, the template will do its best to identify them all, turn any later use of the full expression into an acronym, and add a link to a table at the end. The first use of the expression will have the acronym in parentheses afterwards. The main caveat is that the first mention in the text must be the full expression, the script will not replace an acronym with the expression, because I'm too lazy to manage capitalization edge cases. For example, super useful acronym (SUA).

- Super useful acronym

1.5 ALGORITHMS

Quantum mechanics is not going to be replaced by something that's easier.

— Carlton Caves

In June 1925, Heisenberg fled the Göttingen pollen and beached Helgoland, a collection of orange rocks on the North Sea. He was very productive there, and, in the resulting publication [2], he stated that the product of some physical values, when using the calculation rules of the contemporary theory of quanta, is not commutative — this is recognized by many as the birth of modern quantum mechanics.

In June 2025, a centennial conference was held on the same island to celebrate the result and ponder its consequences. It illuminated the amount of work built on top of this achievement, and simultaneously, the depth of the still unknown. Technological advancement has been remarkable: arrays of atoms can be precisely controlled [3,4]; interactions between gravity and quantum physics are a future milestone rather than a far-off dream [5], and, of special interest to this thesis, quantum computers are, though today not significantly more powerful than classical computers in many regards, a reality [1].

2.1 QUANTUM COMPUTERS

These quantum devices, as foretold in a slightly overused quotation of Richard Feynman, would be used to simulate the dynamical evolution of quantum systems, because “nature isn’t classical”, unfortunately [6]. It has not however been proven that quantum computers can perform computations that classical computers cannot perform [7], but it is widely believed that quantum dynamics cannot, in general, be efficiently simulated on classical computers, and the conjecture is used as a basis for claims of theoretical quantum advantage [8,9]. In fact, if one could efficiently simulate quantum dynamics on a classical device such as a laptop or a supercomputer, then any quantum algorithm could be efficiently run on the device by simulating the quantum computer that runs the program.

If that is not the case, quantum computers have a good chance of exceeding classical capabilities. For that to happen, we must design new quantum algorithms that conform to the constraints of quantum devices, and this requires us to radically change our way of thinking about algorithms. In short, the quantumness of quantum computers must be fully leveraged and understood for them to surpass classical computers: a plane can drive on a road just like a car, but it’s more useful when it flies.

2.2 OUTLOOK

We can only see a short distance ahead, but we can see plenty there that needs to be done.

— Alan Turing

Feynman predicted that computers might become quantum by 2050 [10]. Though he could not predict the difficulties we would meet along the way, it seems that his idea of quantum mechanical computers is slowly turning out to be true, as our NISQ computers turn fault-tolerant, and the algorithms that yesterday

were beyond our grasp are run on today's devices. Looking forward, we hope that quantum digital cooling for fermionic systems beyond toy models becomes attainable, if not commonplace, on tomorrow's devices.

On a more philosophical note, it is startling to see how much of quantum mechanics is still a mystery, and the disagreements on what it tells us about reality are witnesses to its enduring strangeness. Quantum computing helps us lift the veil of this unknown, using a mystery to solve itself; and still, it matters little how advanced quantum devices get: our eyes only see the classical world, and only through it do we get closer to this stranger truth. On Helgoland, Anton Zeilinger said that quantum mechanics links classical data with classical data. To reformulate this in Feynman's words, "nature is classical, dammit!"

BIBLIOGRAPHY

1. **M. A. NIELSEN, I. L. CHUANG** (2010). “Quantum Computation and Quantum Information: 10th Anniversary Edition” Cambridge University Press.
2. **W. HEISENBERG** (1925). “A Quantum-Theoretical Reinterpretation of Kinematic and Mechanical Relations” *Z. Phys.* doi: [10.1007/BF01328377](https://doi.org/10.1007/BF01328377)
3. **M. LUKIN** (2025). “Exploring the Quantum Computing Frontier Using Neutral Atoms” *Helgoland 2025: 100 Years of Quantum Mechanics, Helgoland, Germany*.
4. **S. J. EVERED et al.** (2026). “High-Fidelity Entangling Gates and Nonlocal Circuits with Neutral Atoms” <https://scirate.com/arxiv/2604.25987> (retrieved 4 May 2026)
5. **J. YE** (2025). “Scaling Clock Performance for Fundamental Physics” *Helgoland 2025: 100 Years of Quantum Mechanics, Helgoland, Germany*.
6. **A. TRABESINGER** (2012). “Quantum Simulation” *Nature Physics*. doi: [10.1038/nphys2258](https://doi.org/10.1038/nphys2258)
7. **S. AARONSON** (2009). “BQP and the Polynomial Hierarchy” doi: [10.48550/arXiv.0910.4698](https://doi.org/10.48550/arXiv.0910.4698)
8. **C.-F. CHEN et al.** (2023). “Local Minima in Quantum Systems” doi: [10.48550/arXiv.2309.16596](https://doi.org/10.48550/arXiv.2309.16596)
9. **J. HAAH et al.** (2023). “Quantum Algorithm for Simulating Real Time Evolution of Lattice Hamiltonians” *SIAM Journal on Computing*. doi: [10.1137/18M1231511](https://doi.org/10.1137/18M1231511)
10. **R. P. FEYNMAN** (2018). “Feynman Lectures On Computation” Boca Raton: CRC Press. doi: [10.1201/9780429500442](https://doi.org/10.1201/9780429500442)

APPENDIX

You can also write stuff after you're done with the main text if that's your thing.

A AND THE NUMBERING WILL CHANGE

A.1 AS EXPECTED OF THE APPENDIX

A.2 AND SO ON

A.2.1 AND SO FORTH

B IT SHOULD FOLLOW THE RULES

B.1 AS EXPECTED

ACRONYMS

- **ANAC** another acronym
- **SUA** super useful acronym

COLOPHON

This thesis was written in Typst rather than the usual LaTeX, on a template built *ab initio*. The template was nonetheless strongly inspired by the venerable ClassicThesis LaTeX package.

I used the fonts

- Arial for the cover,
- Tex Gyre Heros for titles,
- Gentium Book Basic for body text,
- New Computer Modern Math for math,
- Roboto Mono for monospace, and
- AuriocusKalligraphicus for chapter numbers and some diagrams.

This document was last compiled on Thursday 06/08/2026, with Typst 0.15.0.